

Forecasting-Based Bias Correction in Difference-in-Differences: Simulation and Empirical Evidence

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Working-draft layout: A4, Times New Roman 12 pt, double-spaced body text, 2.54 cm margins; compact tables and notes. Temporary settings, not an official university template.

Abstract

This thesis evaluates Difference-in-Differences corrections that predict the untreated treated–control gap and subtract its change from a common two-way fixed effects (TWFE) estimate. A simulation compares TWFE, Linear, Synthetic Control, Random Forest, Gradient Boosting and a scenario-matched reference across seven scenarios, with 200 units, 20 periods, true ATT two and 500 replications per scenario. Random Forest reduces RMSE by 22–39% in moderate and stronger linear or quadratic departures, but not under parallel trends, weak divergence or high noise. The implemented Linear correction equals TWFE. Scenario-matched correction reduces average bias further, without always minimizing RMSE.

The framework is applied to homicide counts during the 2020 police strike in Ceará, Brazil. A pre-strike RMSE rule uses 154 rolling 84-day training/13-day placebo windows. Random Forest ranks first for six of seven outcomes. For total homicides its placebo RMSE is 0.00809 versus TWFE’s 0.00830, and its strike-period estimate is 0.03606 versus 0.03437 homicides per municipality/district-day. Repeating selection in 99 AIS draws selects no correction 61 times and Random Forest 15 times; the selected-procedure interval $[-0.0254, 0.1433]$ includes zero. With only five treated AIS, uncertainty is substantial. Historical prediction does not establish removal of causal bias, and the empirical exposure contrast is not the statewide strike effect. The evidence supports conditional, auditable correction rather than a universally preferred method.

1 Introduction

Difference-in-Differences (DiD) provides a way to estimate treatment effects when randomized assignment is unavailable and outcomes are observed before and after an intervention. By comparing the change in outcomes for treated units with the corresponding change for untreated units, it removes permanent differences in outcome levels and shocks shared by the two groups. These features make the design useful for evaluating interventions introduced to some units while other units remain untreated.

Its causal interpretation nevertheless requires a credible account of how treated outcomes would have evolved without treatment. Parallel trends requires the untreated outcome changes of the two groups to agree, in expectation, over the relevant comparison periods. It permits persistent level differences. It does not permit a systematic change in the untreated treated–control gap that is mistaken for a treatment effect. When treatment assignment depends on characteristics that also govern outcome growth, conventional DiD combines the effect of treatment with differential untreated evolution.

This thesis studies the latter component as time-varying selection bias. Examining pre-treatment outcome patterns may reveal divergence, but identifying that divergence does not itself supply an adjusted estimate. A correction also requires an estimate of the untreated group difference after treatment begins and a baseline consistent with the original DiD comparison. The forecasting approach considered here estimates that counterfactual difference and subtracts its predicted change from the conventional estimator.

The research question is whether forecasting-based corrections improve treatment-effect estimation relative to a common two-way fixed effects (TWFE) DID, how much they improve it, and what precision costs accompany the adjustment. Linear, Synthetic Control, Random Forest and Gradient Boosting are evaluated alongside a scenario-matched gap correction that uses the known simulation functional form. This last procedure is a reference for correct specification, not the sole object of evaluation. An empirical application then uses a selection rule based only on pre-treatment predictive performance, without knowing the true untreated gap or ATT.

The simulation contribution is a controlled comparison across seven scenarios with a known average treatment effect on the treated (ATT). The scenarios include parallel trends, increasing

linear trend heterogeneity coupled with treatment selection, quadratic trend heterogeneity, and higher stochastic noise. The panel dimensions and treatment effect are held fixed, allowing the results to be interpreted in relation to the dimensions actually varied. Performance is assessed jointly using bias, variance and root mean squared error (RMSE), with Monte Carlo uncertainty used to qualify close numerical comparisons.

The empirical contribution applies the same adjustment logic to municipality/district-day homicide outcomes during the 2020 police strike in Ceará studied by Mancha [2025]. The correction is deliberately anchored to a standard unit/date TWFE DID estimated from the replication data; it does not adjust Mancha’s published interaction coefficient. No correction, a linear group-gap correction, Synthetic Control, Random Forest and Gradient Boosting are compared in common rolling pre-treatment placebo windows. This converts the simulation’s known-truth evaluation into an implementable pre-treatment-only model-selection exercise.

Random Forest lowers simulation RMSE in S2–S5 but not in the parallel, weak-linear or high-noise scenarios. The scenario-matched reference reduces signed bias further, while incurring enough forecast variance to lose to TWFE in S1 and S6 and to Random Forest in S4. In Ceará, Random Forest has the lowest pre-strike placebo RMSE for six outcomes, but its total-homicide advantage over no correction is only 2.6%. Complete AIS re-selection often changes the winner. The joint evidence therefore separates average-bias reduction from overall accuracy and historical prediction from causal identification.

Section 1.1 positions the research. Section 2 defines the simulated data and scenarios. Section 3 describes the estimand, forecasting procedures and evaluation measures. Sections 4 and 5 report and evaluate the simulation. Section 6 presents the Ceará application. Sections 7 and 8 synthesize the evidence.

1.1 Related literature and research positioning

1.1.1 Identification and departures from parallel trends

Canonical DID imposes restrictions on untreated changes, anticipation and sampling [Roth et al., 2023]. Unit effects remove permanent level selection, not selection on untreated growth. Conditional DID uses observed covariates under conditional identifying restrictions [Abadie,

2005]; doubly robust DID combines propensity-score and outcome models [Sant’Anna and Zhao, 2020]. Neither result implies that prediction using unrelated covariates removes latent trend selection. Here the simulated trend loading is independent of the observed covariates.

Pre-treatment evidence cannot prove the identifying assumption. Pretrend tests may lack power against consequential violations, and conditioning on passing them can distort inference [Roth, 2022]. Researchers must explain why different groups would have comparable untreated evolution [Kahn-Lang and Lang, 2020]. A fitted pre-gap relationship is therefore an extrapolation restriction, not identification of the missing post-treatment outcome.

Responses to nonparallel evolution include sensitivity analysis under restrictions on departures [Rambachan and Roth, 2023] and event-study identification using covariates related to latent confounds [Freyaldenhoven et al., 2019]. Neither is implemented here. This thesis compares specified forecasting procedures under a common correction target, rather than claiming identification under unrestricted nonparallel trends.

The question differs from treatment-timing problems. Goodman-Bacon [2021] examines TWFE comparisons across adoption dates; Sun and Abraham [2021] studies contamination of event-study coefficients by heterogeneous effects. With one adoption date here, the problem is differential untreated gap growth, not already-treated comparisons. Timing-related weighting solutions do not themselves identify that growth.

1.1.2 Counterfactual prediction, donor support and extrapolation

Synthetic Control constructs a counterfactual from weighted untreated donors [Abadie et al., 2010]. Credible application depends on the donor pool and pre-intervention fit, not simply on naming the estimator [Abadie et al., 2015, Abadie, 2021]. The one-mean matching specification retained in this thesis provides a restricted test of that idea: equality of a pre-treatment mean does not ensure equality of latent trend loadings or of the full path. Its performance is consequently evidence about this implementation, not a verdict on all Synthetic Control designs.

Augmented Synthetic Control addresses imperfect fit with a regularized outcome model [Ben-Michael et al., 2021]; matrix completion imputes untreated panel entries using low-rank structure [Athey et al., 2021]. These related but unimplemented approaches establish an ex-

isting history of counterfactual imputation and adjustment. The gap-change identity here is a comparison framework, not a claim of first invention.

Random Forest averages randomized trees [Breiman, 2001]; Gradient Boosting builds an additive predictor through loss-reducing updates [Friedman, 2001]. Nonlinear flexibility within observed support does not specify continuation of a time trend beyond it. Section 3 explains the retained constant-leaf implementations. Prediction and causal identification are different objectives [Mullainathan and Spiess, 2017]: the target here is an untreated aggregate gap change, not an individual outcome label.

1.1.3 Validation, selection and uncertainty

Forecast evaluation should reproduce the information and horizon available when the forecast is used. Rolling-origin validation trains before each forecast origin and evaluates future observations [Hyndman and Athanasopoulos, 2021]. Time-series cross-validation is not governed by a universal rule: Bergmeir et al. [2018] establishes validity for particular autoregressive settings with suitable error conditions. The empirical 84/13-day exercise is chosen to match this thesis’s deployment problem, without assuming those results establish independence of its overlapping training windows. Cawley and Talbot [2010] further shows that finite validation criteria can themselves be overfit during selection. Reporting all candidates and repeating selection in AIS draws makes the observed ranking’s instability visible; it does not supply an independent causal validation sample.

Inference is a separate issue from counterfactual prediction. Serial dependence can severely distort conventional DID inference [Bertrand et al., 2004]; increasing the number of observations within correlated clusters is not equivalent to increasing independent information [Cameron and Miller, 2015]. With few policy-changing groups, Conley and Taber [2011] develops inference using additional restrictions and many nonchanging groups. Ferman and Pinto [2019] addresses heteroskedasticity, including variation associated with group sizes, and MacKinnon and Webb [2017] documents difficulties for bootstrap inference with uneven cluster structures. These results motivate caution about the five treated AIS in this application. The reported stratified percentile bootstrap is not claimed to implement those specialized

methods or to inherit their guarantees.

Known potential outcomes permit ATT-error evaluation in simulation; historical placebo windows permit forecast selection in Ceará. The question is whether correction remains useful after variance costs, counterfactual restrictions and selection uncertainty, not whether an algorithm universally dominates.

2 Simulation Data

2.1 Potential outcomes and treatment assignment

Each replication contains a balanced panel of $N = 200$ units observed over $T = 20$ periods. Periods $1, \dots, 12$ precede treatment and periods $13, \dots, 20$ follow its introduction. Untreated potential outcomes satisfy

$$Y_{it}(0) = \alpha_i + 0.10t + \theta_i f_s(t) + \varepsilon_{it}, \quad i = 1, \dots, 200, \quad t = 1, \dots, 20, \quad (1)$$

where s denotes the scenario, α_i is a unit effect, $0.10t$ is a common time effect, and $\theta_i f_s(t)$ generates heterogeneous trends. The idiosyncratic disturbance is independent Gaussian noise with mean zero and scenario-specific variance $\sigma_{\varepsilon,s}^2$.

Three time-invariant observed covariates are generated independently: $X_{1i} \sim N(0, 1)$, $X_{2i} \sim \text{Bernoulli}(0.5)$ and $X_{3i} \sim U[-1, 1]$. They enter the outcome through

$$\alpha_i = 0.60X_{1i} - 0.40X_{2i} + 0.30X_{3i} + a_i, \quad a_i \sim N(0, 1). \quad (2)$$

The trend loading is $\theta_i \sim N(0, \sigma_{\theta,s}^2)$, independently of these covariates and a_i . In the parallel-trends scenario, $\theta_i = 0$ for every unit. Disturbances are independent across units and periods; the design contains neither serial correlation nor stochastic common shocks.

Treatment-group membership is assigned once for each unit according to

$$D_i \mid \theta_i \sim \text{Bernoulli}(p_i), \quad p_i = \frac{\exp(-0.40 + \delta_{1,s}\theta_i)}{1 + \exp(-0.40 + \delta_{1,s}\theta_i)}. \quad (3)$$

The treated share is random, with both groups required in each panel. All final replications contain both groups on their first generation attempt. Positive $\delta_{1,s}$ makes treatment more likely for units with larger trend loadings. Thus, except in the parallel-trends scenario, treatment selection and heterogeneous trends can generate an increasing untreated group difference.

Let $P_t = \mathbf{1}\{t > 12\}$ and $D_{it} = D_i P_t$. Outcomes are constructed as

$$Y_{it}(1) = Y_{it}(0) + 2P_t, \quad Y_{it} = Y_{it}(0) + 2D_{it}. \quad (4)$$

There is no anticipatory effect. The treatment effect is constant across units and post-treatment periods, so both the period-specific post-treatment ATT and its average over periods 13–20 equal $\tau = 2$. Oracle untreated outcomes are retained for simulation evaluation, but are excluded from the forecasting models’ inputs.

2.2 Scenario variation and Monte Carlo design

Table 1 specifies all seven scenarios. The labels S0–S6 are used consistently in subsequent tables and figures. Shared parameters are the panel dimensions, treatment date, ATT, intercept coefficients, mean trend loading zero, treatment-assignment intercept -0.40 , and common time slope 0.10 .

Table 1: Implemented simulation scenarios.

Scenario	Trend function and description	$\sigma_{\theta,s}$	$\delta_{1,s}$	$\sigma_{\varepsilon,s}$
S0	Linear; parallel trends	0	0	1
S1	Linear; weak heterogeneity	0.0200	10	1
S2	Linear; moderate heterogeneity	0.0500	10	1
S3	Linear; strong heterogeneity	0.1000	10	1
S4	Quadratic; moderate heterogeneity	0.0025	200	1
S5	Quadratic; strong heterogeneity	0.0050	200	1
S6	Linear; moderate heterogeneity, high noise	0.0500	10	3

Linear scenarios use $f_s(t) = t$; quadratic scenarios use $f_s(t) = t^2$. Entries for $\sigma_{\theta,s}$ and $\sigma_{\varepsilon,s}$ are standard deviations, not variances. All scenarios have $N = 200$, $T = 20$, $T_0 = 12$ and $\tau = 2$.

S1–S3 increase the dispersion of linear trend loadings while holding the assignment coefficient fixed. This increases both the scope for heterogeneous growth and the dispersion of treatment propensities. S4–S5 introduce curvature using smaller loading scales and a larger

assignment coefficient. They are not a change in functional form alone relative to the linear scenarios. S6 retains S2’s trend and assignment parameters but triples the error standard deviation, increasing its variance ninefold. Scenario-specific seeds mean that S2 and S6 do not share identical underlying draws.

For intuition, conditional on realized group composition and latent effects, the expected untreated gap is the difference in mean unit effects plus the difference in mean trend loadings multiplied by $f_s(t)$. Differencing the average post and pre gaps removes the unit-effect difference. The post-minus-pre average of t is 10, whereas that of t^2 is $670/3$. These deterministic multipliers explain why small quadratic loadings can still create appreciable DiD bias. Random group-average disturbances contribute sampling variation around this conditional relationship.

The completed Monte Carlo study contains $R = 500$ replications per scenario. All six estimators are evaluated on the same panel within a replication, giving 3,500 panels, 17,500 original estimator results, and 3,500 newly generated scenario-matched estimates. A separate 50-replication pilot is excluded from the reported final summaries. With scenario index $j = 1, \dots, 7$, the final data seed is $20370000 + 100000j + r$. Random Forest and Gradient Boosting use offsets of 3,000,000 and 4,000,000, respectively, from that data seed. The calculations use R; the reproduction environment is R 4.5.0 with Random Forest 4.7-1.2, GBM 2.3.1 and Synth 1.1-10.

3 Methodology

3.1 TWFE and the selection-bias estimand

The identities below are sample accounting relationships. A causal interpretation additionally requires no anticipation, consistency of treatment definitions, and a defensible untreated comparison path [Roth et al., 2023]. Predicting that path replaces exact parallel trends with the candidate’s forecasting restriction; it does not remove the need for such a restriction. Write $\bar{Y}_{dt}$ for the sample mean outcome among units with $D_i = d$ at time t , and define $G_t = \bar{Y}_{1t} - \bar{Y}_{0t}$. For any time series z_t , let $\bar{z}_{\text{pre}}$ and $\bar{z}_{\text{post}}$ denote its equally weighted averages over the 12 pre-treatment and eight post-treatment periods, respectively. The conventional TWFE specification

is

$$Y_{it} = a_i + \lambda_t + \tau^{\text{TWFE}} D_{it} + u_{it}. \quad (5)$$

With this balanced panel and common adoption date, its estimated treatment coefficient is exactly

$$\hat{\tau}^{\text{TWFE}} = \bar{G}_{\text{post}} - \bar{G}_{\text{pre}}. \quad (6)$$

The Monte Carlo study uses this algebraic representation. It evaluates a single average post-treatment estimate per replication. Time-invariant covariates are absorbed by unit effects and do not add separately identified regressors to this benchmark.

Define $G_t^0 = \overline{Y(0)}_{1t} - \overline{Y(0)}_{0t}$. The realized selection-bias component aligned with Equation 6 is

$$B = \bar{G}_{\text{post}}^0 - \bar{G}_{\text{pre}}^0, \quad \hat{\tau}^{\text{TWFE}} = \tau + B. \quad (7)$$

The population counterpart of B captures systematic nonparallel evolution. Its realized value additionally includes differences in group-average disturbances. Under parallel trends, its expectation is zero, although it need not vanish in a particular replication. The baseline is the average pre-treatment gap; replacing it by the last pre-period gap would change the quantity required to adjust this TWFE estimator.

3.2 Forecasting-based adjustment

For forecasting method m , let $\hat{G}_{t,m}^0$ denote its predicted untreated treated-control gap. The period-specific correction component and its post-treatment average are

$$\widehat{SB}_{t,m} = \hat{G}_{t,m}^0 - \bar{G}_{\text{pre}}, \quad \hat{B}_m = \frac{1}{8} \sum_{t=13}^{20} \widehat{SB}_{t,m}. \quad (8)$$

The evaluated adjusted estimator is

$$\hat{\tau}_m^{\text{adj}} = \hat{\tau}^{\text{TWFE}} - \hat{B}_m. \quad (9)$$

Thus an untreated outcome prediction is an input to a predicted group gap; the change in that gap is an estimated selection-bias term; and subtracting that term yields a treatment-effect estimate. These quantities have distinct roles.

For Linear, Random Forest and Gradient Boosting, the predicted gap is the difference between predicted untreated outcomes averaged separately over both groups. All three methods train exclusively on the 2,400 observations from periods 1–12, then generate predictions for all units and periods. Synthetic Control instead forecasts the aggregate treated path from weighted observed donor outcomes and subtracts the observed control-group mean. It therefore uses contemporaneous untreated donor outcomes in post-treatment periods. No method fits to treated post-treatment outcomes or to the stored oracle untreated outcomes.

3.3 Scenario-matched group-gap correction

The scenario-matched reference models the twelve observed pre-treatment gaps directly. Let $g_t = \bar{Y}_{1t} - \bar{Y}_{0t}$ for $t \leq 12$. In S0, where the DGP imposes parallel trends, no correction is applied and $\hat{B} = 0$. In the linear-violation scenarios S1–S3 and S6, ordinary least squares estimates

$$g_t = a + bt + e_t, \quad t = 1, \dots, 12. \quad (10)$$

In the quadratic scenarios S4–S5 it estimates

$$g_t = a + bt + ct^2 + e_t. \quad (11)$$

Each fitted polynomial is extrapolated to periods 13–20. The correction remains the difference between the mean predicted post-treatment gap and the observed mean pre-treatment gap, and the adjusted estimate remains Equation 9. Because OLS with an intercept reproduces the mean of the dependent variable in sample,

$$\hat{B}_{\text{linear}} = 10\hat{b}, \quad \hat{B}_{\text{quadratic}} = 10\hat{b} + \frac{670}{3}\hat{c}. \quad (12)$$

The constants follow from $\bar{t}_{\text{post}} - \bar{t}_{\text{pre}} = 10$ and $\bar{t}_{\text{post}}^2 - \bar{t}_{\text{pre}}^2 = 670/3$.

The polynomial order is assigned from the known simulation scenario. The method therefore measures performance when the correction’s functional form is correctly specified; it is an oracle benchmark, not a deployable data-driven selection rule. In an empirical application, the order would need to be specified before examining post-treatment treated outcomes or chosen only from pre-treatment evidence with suitable validation and uncertainty accounting. This distinction prevents the simulation’s DGP knowledge from being treated as information available to an applied researcher.

3.4 Common-slope linear forecasting

The linear forecaster includes unit intercepts and a single common deterministic time slope:

$$Y_{it} = c_i + \beta t + v_{it}, \quad t \leq 12, \quad \hat{Y}_{it}(0) = \hat{c}_i + \hat{\beta}t. \quad (13)$$

The slope is estimated from within-unit deviations in the pre-treatment sample. Each fitted intercept is $\hat{c}_i = \bar{Y}_{i,\text{pre}} - \hat{\beta} \bar{t}_{\text{pre}}$. The time-invariant covariates are absorbed by these intercepts. There are no group-specific or unit-specific slope terms.

Because both groups have the same observed periods, the common forecast slope cancels in their predicted difference. The remaining difference in fitted intercepts equals $\bar{G}_{\text{pre}}$. Consequently,

$$\hat{G}_{t,\text{Linear}}^0 = \bar{G}_{\text{pre}}, \quad \hat{B}_{\text{Linear}} = 0, \quad \hat{\tau}_{\text{Linear}}^{\text{adj}} = \hat{\tau}^{\text{TWFE}}. \quad (14)$$

This identity is part of the implemented comparison and is retained in reporting the results.

3.5 Synthetic Control

Synthetic Control represents the treated group by its average outcome series and uses every untreated unit as a donor. The sole matching predictor is each unit’s mean outcome over periods 1–12. No separate lagged outcomes, covariates or trend measures are included. With

nonnegative donor weights summing to one, the relevant matching problem is

$$\min_{\{w_j\}} \left(\bar{Y}_{1,\text{pre}} - \sum_{j:D_j=0} w_j \bar{Y}_{j,\text{pre}} \right)^2, \quad w_j \geq 0, \quad \sum_{j:D_j=0} w_j = 1. \quad (15)$$

Scaling the single predictor does not change this objective’s minimizers. The Synth implementation fixes the predictor weight to one in this case and obtains donor weights using its default constrained quadratic-programming routine. Although the pre-treatment periods are also supplied as outcome-fit periods, there is no outer predictor-weight optimization with only one predictor. The full trajectory loss is calculated after the weights are obtained; it does not select them in this specification.

The resulting forecasts are

$$\widehat{\bar{Y}(0)}_{1t,\text{SC}} = \sum_{j:D_j=0} \hat{w}_j Y_{jt}, \quad \hat{G}_{t,\text{SC}}^0 = \sum_{j:D_j=0} \hat{w}_j Y_{jt} - \bar{Y}_{0t}. \quad (16)$$

Substitution into Equation 9 yields the average post-treatment difference between observed treated outcomes and the synthetic series. The comparison does not include an additional adjustment for the synthetic series’ pre-treatment residual gap.

3.6 Random Forest and Gradient Boosting

Random Forest uses six predictors: numeric unit identifier, time, treatment-group membership, and the three observed covariates. It fits a regression forest with 500 trees and two candidate predictors per split. Other settings retain the installed regression defaults, including replacement sampling and terminal node size five. Unit identifiers are numerical predictors, rather than a complete set of categorical fixed effects. Predictions for both groups enter Equation 8. Hyperparameters are fixed in advance; no cross-validation search is undertaken.

Gradient Boosting uses the same substantive predictors, ordered as time, treatment-group membership, the three covariates and numeric unit identifier. It minimizes Gaussian loss using 800 trees of interaction depth three, shrinkage 0.01, minimum node size ten and a bag fraction of 0.7. The full pre-treatment sample is available for training, cross-validation folds are zero,

and prediction uses all 800 trees. There is no validation-based stopping rule or hyperparameter selection.

Both procedures use constant-leaf regression trees. For a given unit, every post-treatment time lies beyond the largest training time, while all remaining predictors are constant. Each unit therefore follows the same branches at every post-treatment date and receives the same prediction throughout periods 13–20. The methods can learn nonlinear relationships and group differences within the training sample, but these specifications do not extrapolate a continuing time trend beyond it.

3.7 Transfer to the empirical design

The empirical application preserves Equation 9 but cannot copy the oracle functional-form choice or synthetic covariates. All five candidates share a balanced unit/date TWFE, unit weights and the 84-day baseline. Donor forecasts may use contemporaneous controls; tree forecasts use pre-strike outcomes and forecast-known calendar/group indicators, fitting arm-level rather than unit-level outcomes. These adaptations are specified in Section 6. With the untreated strike gap unobserved, the application evaluates forecasts in earlier 84/13-day windows, selects by placebo RMSE, and reports all final estimates with AIS resampling. This is historical selection evidence, not observed strike-period ATT bias.

3.8 Performance metrics and Monte Carlo uncertainty

Let $\hat{\tau}_{r,s,m}$ denote a treatment-effect estimate and $e_{r,s,m} = \hat{\tau}_{r,s,m} - 2$. Omitting scenario and method subscripts for brevity, the signed Monte Carlo bias and finite-sample empirical variance are

$$\widehat{\text{Bias}} = \frac{1}{R} \sum_{r=1}^R e_r, \quad V_R = \frac{1}{R} \sum_{r=1}^R (\hat{\tau}_r - \bar{\hat{\tau}})^2. \quad (17)$$

Absolute bias is the absolute value of signed bias. It differs from mean absolute error, which averages the absolute replication errors. RMSE is

$$\widehat{\text{RMSE}} = \left(\frac{1}{R} \sum_{r=1}^R e_r^2 \right)^{1/2}, \quad \widehat{\text{RMSE}}^2 = \widehat{\text{Bias}}^2 + V_R. \quad (18)$$

The stored empirical standard deviation uses denominator $R - 1$. The variance in Equation 17 is reconstructed with denominator R for the exact finite-sample decomposition in Equation 18; both conventions are retained in the supplementary numerical results. Relative signed bias divides signed bias by two and multiplies by 100. Additional recorded summaries include mean estimates, failure rates, selection-bias errors and computation times. No single primary performance metric is designated.

Monte Carlo standard error (MCSE) for an estimated mean or signed bias is

$$\widehat{\text{MCSE}} = \frac{s_{\hat{\tau}}}{\sqrt{R}}, \quad s_{\hat{\tau}}^2 = \frac{1}{R-1} \sum_{r=1}^R (\hat{\tau}_r - \bar{\hat{\tau}})^2. \quad (19)$$

To assess close comparisons, supplementary diagnostics pair methods within the same replication. For methods m and k , define $q_r = e_{r,m}^2 - e_{r,k}^2$. The mean of q_r estimates their difference in mean squared error, and its MCSE is the sample standard deviation of q_r divided by $\sqrt{R}$. This diagnostic accounts for common simulated panels and supplements the original metrics without imposing a primary ranking criterion.

MCSE quantifies uncertainty in an estimated performance measure, not a within-dataset causal-effect interval [Morris et al., 2019]. Dataset-specific ATT interval coverage is not evaluated and is not reported as a finding.

Finally, the selection-bias forecast error and adjusted treatment-effect error satisfy

$$\hat{\tau}_m^{\text{adj}} - \tau = B - \hat{B}_m. \quad (20)$$

Their RMSEs are identical. This identity makes accurate estimation of the relevant aggregate bias directly consequential for treatment-effect error, but it does not establish that low individual-outcome prediction error guarantees low ATT error. No repeated outcome-level forecast-validation metric is available for the final Monte Carlo comparison.

4 Simulation Results

4.1 Validation and numerical presentation

The archived scenario-matched calculation regenerated all 3,500 final panels using the original seeds; its TWFE and realized bias match the original results within 4.9×10^{-15} . This revision re-summarizes those stored estimates without refitting models. All estimates are finite, with 500 replications per scenario. Equations 12 and 20, and the finite-sample MSE decomposition, hold to numerical precision.

Table 2 supplies mean ATT estimates, signed bias, RMSE and variance for every scenario and method. Absolute bias is the magnitude of the signed-bias column; standard deviations, Monte Carlo errors and correction diagnostics are retained in the accompanying numerical files. Section 5 interprets this common result table using MSE decomposition and paired-error diagnostics, rather than duplicating its ranking in another summary plot.

Table 2: Updated Monte Carlo performance across six estimators.

Scenario	Estimator	Mean	Bias	RMSE	Variance
S0	Matched gap adj.	1.9942	-0.0058	0.0651	0.00421
	TWFE	1.9942	-0.0058	0.0651	0.00421
	Common-slope adj.	1.9942	-0.0058	0.0651	0.00421
	SC adj.	1.9954	-0.0046	0.0645	0.00413
	RF adj.	1.9986	-0.0014	0.0965	0.00930
	GBM adj.	1.9966	-0.0034	0.0787	0.00619
S1	Matched gap adj.	1.9965	-0.0035	0.1372	0.01880
	TWFE	2.0406	0.0406	0.0819	0.00505
	Common-slope adj.	2.0406	0.0406	0.0819	0.00505
	SC adj.	2.0413	0.0413	0.0822	0.00505
	RF adj.	2.0273	0.0273	0.0988	0.00902
	GBM adj.	2.0476	0.0476	0.1024	0.00823
S2	Matched gap adj.	2.0002	0.0002	0.1344	0.01805
	TWFE	2.2280	0.2280	0.2486	0.00980
	Common-slope adj.	2.2280	0.2280	0.2486	0.00980
	SC adj.	2.2221	0.2221	0.2428	0.00964
	RF adj.	2.1443	0.1443	0.1781	0.01090
	GBM adj.	2.2522	0.2522	0.2736	0.01127
S3	Matched gap adj.	1.9955	-0.0045	0.1377	0.01893
	TWFE	2.8222	0.8222	0.8345	0.02032
	Common-slope adj.	2.8222	0.8222	0.8345	0.02032

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Table 2 continued

Scenario	Estimator	Mean	Bias	RMSE	Variance
	SC adj.	2.6914	0.6914	0.7089	0.02459
	RF adj.	2.4942	0.4942	0.5088	0.01469
	GBM adj.	2.6997	0.6997	0.7093	0.01343
S4	Matched gap adj.	1.9686	-0.0314	0.3733	0.13836
	TWFE	2.2647	0.2647	0.2836	0.01039
	Common-slope adj.	2.2647	0.2647	0.2836	0.01039
	SC adj.	2.2652	0.2652	0.2841	0.01040
	RF adj.	2.1918	0.1918	0.2217	0.01235
	GBM adj.	2.2747	0.2747	0.2972	0.01284
S5	Matched gap adj.	1.9977	-0.0023	0.4058	0.16469
	TWFE	2.9194	0.9194	0.9314	0.02214
	Common-slope adj.	2.9194	0.9194	0.9314	0.02214
	SC adj.	2.8927	0.8927	0.9054	0.02290
	RF adj.	2.6692	0.6692	0.6827	0.01829
	GBM adj.	2.8761	0.8761	0.8860	0.01741
S6	Matched gap adj.	2.0012	0.0012	0.3971	0.15772
	TWFE	2.2295	0.2295	0.3052	0.04047
	Common-slope adj.	2.2295	0.2295	0.3052	0.04047
	SC adj.	2.2702	0.2702	0.3338	0.03844
	RF adj.	2.1380	0.1380	0.3205	0.08367
	GBM adj.	2.2472	0.2472	0.3184	0.04025

5 Evaluation

This section evaluates the six procedures using the results in Section 4. The comparison concerns accuracy of an ATT estimate with truth $\tau = 2$, not generic prediction performance. Linear, SC, RF and GB denote the implemented adjustments; Matched denotes the scenario-matched reference. All comparisons are within the same simulated panel. Numerical diagnostics below summarize the saved 500 replications in each scenario and do not involve new model fitting.

5.1 Bias Reduction and Accuracy of ATT Estimates

Table 2 distinguishes substantial removal of systematic overestimation from merely moving the mean closer to two. TWFE's signed bias grows from 0.0406 in S1 to 0.2280 in S2 and 0.8222 in S3. Linear leaves every estimate unchanged, so it provides no bias reduction in any scenario. This is a consequence of the common-slope specification already established in Methodology,

not a finding that all linear corrections are ineffective.

RF reduces signed and absolute bias in every violation scenario, but leaves substantial upward error. In S1–S3 it removes approximately one-third to two-fifths of TWFE’s systematic error; the remaining biases are 0.0273, 0.1443 and 0.4942. Partial rather than complete correction also characterizes the quadratic and high-noise cases.

SC’s bias improvement is most appreciable in S3, from 0.8222 to 0.6914; in S6 bias increases to 0.2702. GB reduces the larger biases in S3 and S5 but increases bias in S1, S2, S4 and S6. Neither retained procedure uniformly corrects differential growth.

In S0 all signed biases are small, ranging from -0.0058 for TWFE to -0.0014 for RF. The draws do not show substantial new systematic bias under parallel trends. Differences of a few thousandths, with bias MCSEs about 0.0029–0.0043, offer little basis for ranking methods without considering precision.

Matched has the smallest observed absolute bias in S1–S6, ranging from 0.0002 to 0.0314, and equals TWFE in S0 by construction. In S4 its mean estimate is 1.9686 rather than TWFE’s 2.2647. The change of sign from positive to negative is not, by itself, evidence of an important overcorrection: the absolute error of the mean falls from 0.2647 to 0.0314, and the latter has MCSE 0.0167. The benefit must still be judged against dispersion, not against the sign alone.

5.2 Precision and Stability of Adjusted Estimates

Small average bias does not imply stability in one panel. Linear has TWFE’s dispersion. SC’s standard deviation is usually similar, but rises from 0.1427 to 0.1570 in S3. RF is noisier in S1 (0.0951 versus 0.0711), yet less dispersed in S3 and S5 (0.1213 and 0.1354 versus 0.1427 and 0.1489). GB also has lower dispersion in those stronger scenarios but higher dispersion in S0–S2 and S4. A variance penalty is not universal.

S2 and S6 isolate a useful noise comparison because they share trend and assignment parameters while the disturbance standard deviation rises from one to three. TWFE’s standard deviation rises from 0.0991 to 0.2014. RF’s rises from 0.1045 to 0.2895 and Matched’s from 0.1345 to 0.3975. RF continues to reduce average bias, and Matched remains nearly unbiased on average, but both produce much more dispersed individual estimates. Their loss of RMSE

advantage is therefore not explained by failure to move the mean toward truth. The scenarios use separate seeds, so this is a comparison of scenario distributions, not paired S2/S6 noise realizations.

Matched’s quadratic corrections have standard deviations 0.3723 in S4 and 0.4062 in S5. Correct form reduces average bias but estimating a continuing shape from twelve pre-period gaps adds extrapolation uncertainty, as Figure 1 shows.

5.3 Overall Estimation Accuracy Relative to TWFE

The exact finite-sample decomposition $MSE = Bias^2 + V_R$ uses variance with denominator R . It prevents a method with smaller bias but much larger variance from being classified as superior on bias alone. Figure 1 separates the two components. RF lowers RMSE by 28.4%, 39.0%, 21.8% and 26.7% in S2–S5, respectively. Yet it raises RMSE by 48.1% in S0, 20.7% in S1 and 5.0% in S6. GB reduces RMSE in S3 and S5, while its remaining scenario results are worse than TWFE. SC’s improvements are smaller except in S3; Linear’s change is zero throughout.

The estimator with the smallest bias is not necessarily the estimator with the lowest RMSE. Matched’s RMSE reductions in S2, S3 and S5 are 46.0%, 83.5% and 56.4%, but in S4 RF has lower RMSE (0.2217) than the nearly unbiased matched quadratic correction (0.3733). In S1 and S6, unadjusted TWFE has the lowest RMSE. In S0, SC has the numerical minimum (0.0645 versus TWFE’s 0.0651), but that close comparison requires Monte Carlo qualification.

Following the distinction between estimator performance and Monte Carlo uncertainty emphasized by Morris et al. [2019], Figure 2 uses squared-error differences paired within replications. SC’s S0 difference from TWFE is -0.0000869 with MCSE 0.0000791, so this slight numerical lead is not clearly resolved. In S3, SC and GB both materially improve on TWFE, but their direct paired MSE difference is only -0.000474 with MCSE 0.00635. Their mutual ordering is likewise unresolved. RF’s reductions relative to TWFE in S2–S5 are large relative to their MCSEs: for example, S2 gives -0.03009 with MCSE 0.00160. Its S6 deterioration is 0.00958 with MCSE 0.00552, so the estimated disadvantage is less clearly resolved than its advantages in S2–S5. These are pointwise Monte Carlo diagnostics, not multiplicity-adjusted



Figure 1: MSE decomposed into squared signed bias and population variance for all six procedures. Vertical scales vary by scenario. The figure diagnoses the sources of estimation error rather than repeating the RMSE ranking.

claims or causal-effect confidence intervals.

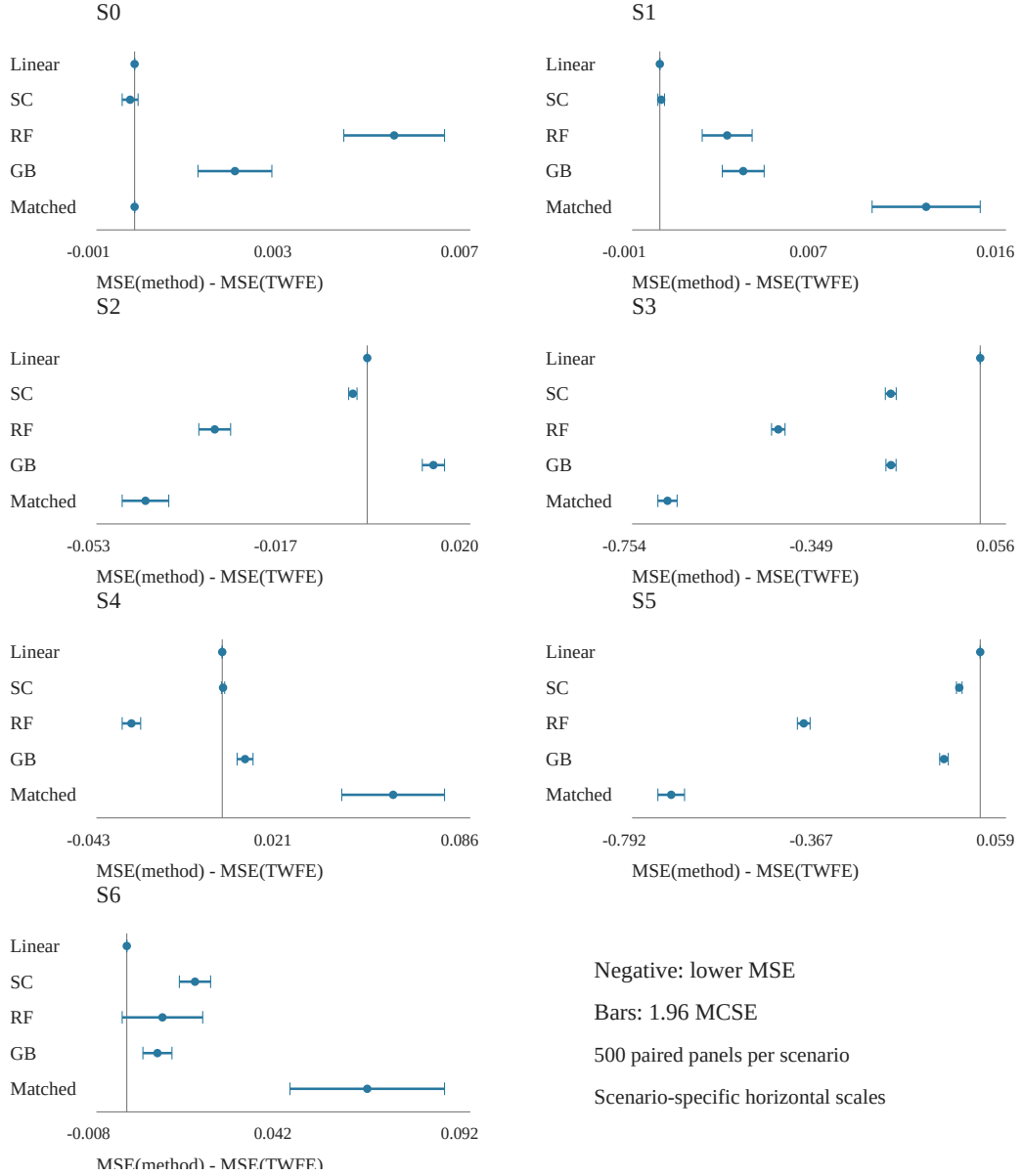


Figure 2: Paired MSE differences from TWFE for every adjustment. Negative values favor adjustment; bars are 1.96 MCSE. Horizontal scales vary by scenario. Linear is zero up to rounding; Matched equals TWFE in S0.

5.4 Accuracy of the Estimated Bias Correction

The target is the realized untreated gap change B , not individual outcome prediction. Table 3 contrasts its mean with estimated corrections. Linear estimates zero; RF generally estimates a positive but insufficient correction. In S3, $\bar{B} = 0.8222$ and $\bar{\hat{B}}_{\text{RF}} = 0.3280$, leaving ATT bias 0.4942. Negative mean corrections by SC or GB can increase upward TWFE bias. Matched

tracks mean B more closely but remains variable.

Table 3: Mean realized and estimated gap-change corrections.

Scenario	$\bar{B}$	Linear	SC	RF	GB	Matched
S0	-0.0058	-0.0000	-0.0012	-0.0045	-0.0024	0.0000
S1	0.0406	-0.0000	-0.0007	0.0133	-0.0070	0.0441
S2	0.2280	-0.0000	0.0059	0.0838	-0.0241	0.2278
S3	0.8222	0.0000	0.1308	0.3280	0.1225	0.8267
S4	0.2647	0.0000	-0.0005	0.0729	-0.0100	0.2961
S5	0.9194	0.0000	0.0268	0.2502	0.0433	0.9217
S6	0.2295	0.0000	-0.0407	0.0915	-0.0177	0.2283

Each entry averages 500 replications. TWFE uses a zero correction. Positive $B - \hat{B}$ implies residual upward ATT error. Individual correction errors and paired uncertainty are supplied in the numerical archive.

Mean correction error is $\overline{\hat{B} - B}$, the negative of signed ATT bias. Its RMSE is exactly the ATT RMSE because $\hat{\tau}^{\text{adj}} - 2 = B - \hat{B}$ in every saved replication. For example, RF’s S3 mean correction error is -0.4942 and correction RMSE is 0.5088 ; Matched’s S4 mean correction error is 0.0314 but correction RMSE is 0.3733 . These diagnose insufficient average correction and noisy correction, respectively. They are alternative interpretations of the same estimation error, not two independent validations. The full correction-error summaries are supplied with the calculation files. No individual-outcome prediction accuracy claim is added because those validation metrics were not recorded for this comparison.

5.5 Performance Across Scenarios and Implications for Use

Three conditional judgments follow. Under parallel trends or weak divergence, correction can add avoidable variation; retaining no correction as a candidate is important. Under moderate and strong divergence at the lower noise level, RF partially improves overall accuracy, and knowledge of the correct linear or quadratic form can improve it further in S2, S3 and S5. Under high noise, average-bias improvement does not ensure lower total error. SC and GB gains depend on the scenario and the particular information and matching specifications. These are complete-procedure comparisons, not algorithm rankings holding inputs constant.

Matched uses the known scenario to choose polynomial order. Its success cannot establish that an applied researcher can select that order from a short pre-period. Nor is it a theoretical

bound on attainable accuracy: RF already has lower RMSE in S4 despite larger bias. It is a reference for correct functional form within the retained design, not a universal recommendation.

The conclusions are conditional on $N = 200$, $T = 20$, twelve pre-periods, eight post-periods, constant ATT two, independent Gaussian errors and the specified selection mechanism. The design does not evaluate serial dependence, heterogeneous or staggered effects, structural breaks, missingness or within-dataset confidence-interval coverage. These remain unassessed questions. The empirical study consequently uses pre-period placebo RMSE to choose among observable procedures and reports all candidates, rather than carrying oracle scenario information or simulation accuracy claims into an unobserved strike counterfactual.

6 Empirical Study

6.1 Institutional setting, source audit, and analysis sample

The empirical application re-examines the 18 February–1 March 2020 police strike in Ceará studied by Mancha [2025]. The primary source is the 27,830 victim records in `Painel.xlsx`; `Gang_Turfs.xlsx` supplies the gang-exposure measure. The released `Daily_Homicides_CE.dta` is an intermediate aggregation of the victim records, not independent evidence. We reconstruct its outcomes from the spreadsheet and compare unit–day totals after identical AIS and municipality normalization. The intermediate file has one duplicated unit–day; after re-aggregation its 27,418 valid-AIS homicides and all unit–day outcome counts agree with the source reconstruction. Seven exact duplicate source rows are retained because the data are victim records and the source does not establish that they are duplicate victims.

We exclude 412 records with absent or non-geographic AIS labels, retain 358 municipality/district units in 22 AIS, and balance every unit over 2,557 dates (1 January 2014–31 December 2020). Of 915,406 unit–days, 893,515 have no recorded homicide and are filled with zero. The main 97-day window has 34,726 unit–days (33,943 zero), 989 victim records, 94 treated units and 264 controls. The seven outcomes are total homicides, men, women, and the released `crime_1`–`crime_4` indicators, all as unit–day counts. To reproduce the released Stata

coding, women equal total minus men; 24 records with unclassified sex therefore enter this residual category. This is a measurement limitation, not an assertion that their victims were female. The original spreadsheet, intermediate file, source-code hash, row flow, and discrepancy checks are preserved in the accompanying reproduction archive.

The primary treated set is AIS 02, 06, 11, 12, and 13, matching the paper’s stated five high-exposure areas. The released Stata code lowercases AIS labels and later tests the uppercase literal AIS 13; that test fails, making its executed treated set AIS 02, 06, 11, and 12. We use this four-AIS version only as a replication and sensitivity case. A separate boundary issue arises from the 75th percentile of `ExposureDistricts`: its value is 0.587728, whereas AIS 07 is 0.588235. A mechanical “greater than percentile” rule includes AIS 07 and yields six areas. We report that six-AIS case separately rather than silently redefine the paper’s intended five areas.

The intervention is statewide, whereas D_i distinguishes high gang exposure. The control arm is therefore not a group of municipalities outside the strike. The estimand is the strike-period change in the high-minus-lower-exposure gap relative to a forecast of its no-strike evolution. Interpreting it causally requires that this forecast capture the relevant differential untreated change and that other differential shocks not be attributed to the strike. It does not identify the statewide total effect of withdrawing police surveillance. The contrast follows the exposure-based comparison in Mancha [2025], with the baseline, sample and weights specified below.

6.2 Common TWFE benchmark and gap correction

The analysis window consists of the 84 days ending 17 February 2020 and the 13 strike days. Each municipality/district receives equal weight within its treatment arm. The common benchmark is the coefficient on $D_i \times S_t$ in

$$Y_{it} = \alpha_i + \lambda_t + \tau^{\text{TWFE}} D_i S_t + u_{it}, \quad (21)$$

where α_i and λ_t are unit and date effects. With the balanced panel and one common treatment date, direct two-way residualization verifies that this coefficient equals the post-strike treated-

minus-control gap less the pre-strike mean gap for all seven outcomes; the largest numerical discrepancy is 6.8×10^{-15} .

Let G_t denote that daily unit-weighted group gap. For each candidate m we use the *same* $\hat{\tau}^{\text{TWFE}}$ and pre-period baseline:

$$\hat{B}_m = \overline{\hat{G}_{m,\text{post}}^0} - \overline{G}_{\text{pre}}, \quad \hat{\tau}_m = \hat{\tau}^{\text{TWFE}} - \hat{B}_m = \overline{G}_{\text{post}} - \overline{\hat{G}_{m,\text{post}}^0}. \quad (22)$$

Every forecast is averaged over seven days in the first strike block and six in the second, equivalently weighting the two block means by 7/13 and 6/13. The no-correction forecast sets $\overline{\hat{G}^0} = \overline{G}_{\text{pre}}$. The linear-gap method fits an ordinary line to twelve pre-strike weekly gap means and extrapolates weeks 13 and 14. Synthetic Control matches the treated arm’s single 84-day pre-mean with nonnegative control-unit donor weights summing to one, then uses contemporaneous donor outcomes to forecast the treated mean and subtracts the contemporaneous unweighted control mean. Because a single matching predictor leaves many exact solutions, this implementation uses a maximum-entropy tie-break based on donor multiplicity; it is a transparent empirical implementation of the simulation’s one-predictor design, not a claim to reproduce every numerical detail of the original Synth solver.

The Random Forest and Gradient Boosting retain the simulation’s tree counts and major tuning choices (500 trees and two candidate features per forest split; 800 depth-three boosting trees, learning rate 0.01, leaf size ten, subsample 0.7). Their empirical prediction target is different. The simulated X_1, X_2, X_3 do not exist in these data, so each model fits the two daily *group mean* series, using only relative day, treatment-arm label, weekday and calendar month. It predicts both group means and takes their difference. Relative day, weekday, month, and arm label are known at forecast time. Models are refitted inside each window. Synthetic Control may use contemporaneous control outcomes; no candidate uses strike-period treated outcomes to fit or select its counterfactual. With constant terminal regions, neither fitted tree procedure extends a time slope beyond the training support. Forest predictions average training responses; boosting adds fitted tree contributions and need not remain within the range of the original responses [Breiman, 2001, Friedman, 2001].

6.3 Pre-strike evaluation and selection

The true untreated strike-period gap is unobserved, so empirical ATT bias cannot be measured. We instead run the same 84-day training/13-day testing exercise at 154 pre-strike origins, moving the origin by 14 days each time. Each origin re-estimates all five candidates and treats the ensuing 13 days as a zero-effect placebo. The resulting pseudo-effect is $\overline{G}_{\text{test}} - \overline{\widehat{G}_{m,\text{test}}^0}$, algebraically the common placebo TWFE less the candidate’s predicted bias term. We report mean signed error, population variance, MAE, RMSE, available windows, and failures. The winner for each outcome is fixed by the smallest pre-strike placebo RMSE, with no inspection of strike outcomes. The 13-day test blocks do not overlap at a 14-day step, but adjacent 84-day training blocks do; paired window differences are correlated and a naive window-level standard error is not an independent-sample uncertainty measure.

Table 4: Total-homicide rolling pre-strike placebo performance (154 windows, 84-day training and 13-day test). The pseudo-effect is zero; variance is the population variance of placebo estimates. SC–entropy uses an entropy tie-break; RF–empirical and GB–empirical use forecast-known calendar and group features.

Method	Signed error	Variance	MAE	RMSE	Windows	Failures
No correction	0.00028	0.000069	0.00675	0.00830	154	0
Linear gap	-0.00008	0.000090	0.00755	0.00946	154	0
SC–entropy	0.00105	0.000076	0.00718	0.00876	154	0
RF–empirical	0.00202	0.000061	0.00642	0.00809	154	0
GB–empirical	0.00018	0.000067	0.00655	0.00820	154	0

For total homicides, Random Forest has the smallest pre-strike RMSE (0.00809), versus 0.00830 for no correction, 0.00820 for Gradient Boosting, 0.00876 for Synthetic Control, and 0.00946 for the linear gap. Its 2.6% advantage over unadjusted TWFE is modest, and it has a positive mean signed placebo error (0.00202), so this ranking is evidence of relative pre-period prediction, not proof that strike-period selection bias was eliminated. A lag-six HAC sensitivity for the paired forest-versus-TWFE MSE difference gives -3.49×10^{-6} with a standard error of 6.42×10^{-6} ; it does not resolve the close ranking statistically. Random Forest also ranks first for men, women and three of the four crime indicators; no correction ranks first for `crime_4`. The full all-outcome performance table is in Table 9 below.

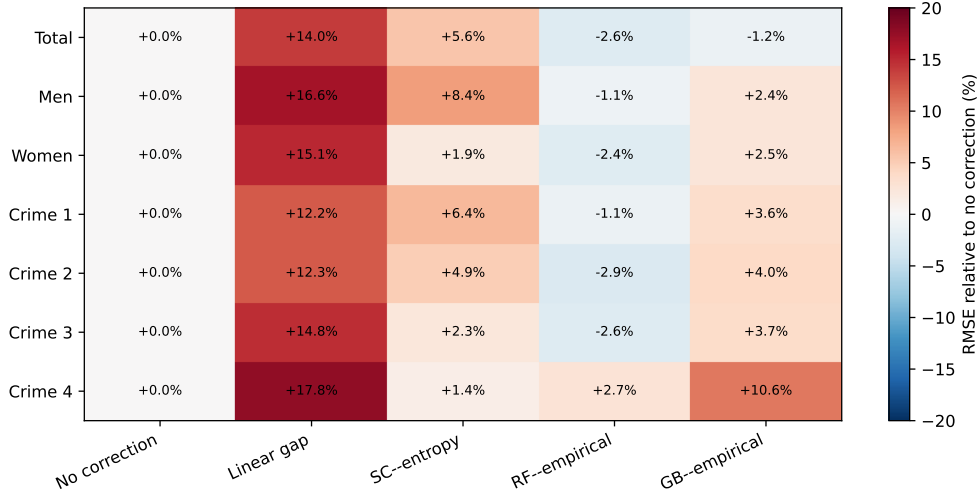


Figure 3: Pre-strike placebo RMSE relative to no correction; negative values indicate lower RMSE. All comparisons use the same 154 rolling windows.

6.4 Strike-period estimates and AIS resampling

After the selection rule is frozen, all five methods are refitted on the actual 84 pre-strike days, and every pre-specified estimate is reported. For total homicides the common TWFE DID is 0.03437 per municipality/district-day. The linear, Synthetic Control, Random Forest, and Gradient Boosting predicted bias terms are respectively 0.00732, 0.04560, -0.00169 , and -0.00136 , giving adjusted estimates 0.02705, -0.01123 , 0.03606, and 0.03573. Synthetic Control’s larger downward adjustment reflects its donor-based forecast of a higher untreated strike-period gap; its information set differs because the control outcomes are contemporaneous. It was not selected by the pre-strike rule. Table 5 gives every outcome, the identical within-outcome TWFE, its candidate-specific bias term, corrected estimate and fixed-method interval.

Table 5: Strike-period TWFE, predicted bias and correction on the common unit/date panel. Percentile intervals resample AIS within treatment arms; * marks the pre-strike RMSE winner.

Outcome	Method	TWFE	Bias $\hat{B}_m$	Adjusted	95% interval	Selected
Total	No correction	0.0344	0.0000	0.0344	[-0.016, 0.142]	
Total	Linear gap	0.0344	0.0073	0.0271	[-0.018, 0.135]	
Total	SC-entropy	0.0344	0.0456	-0.0112	[-0.045, 0.065]	
Total	RF-empirical	0.0344	-0.0017	0.0361	[-0.013, 0.155]	*
Total	GB-empirical	0.0344	-0.0014	0.0357	[-0.012, 0.156]	
Men	No correction	0.0320	0.0000	0.0320	[-0.015, 0.144]	

Outcome	Method	TWFE	Bias	Adjusted	95% interval	Selected
Men	Linear gap	0.0320	0.0062	0.0257	[-0.017, 0.135]	
Men	SC-entropy	0.0320	0.0416	-0.0096	[-0.039, 0.067]	
Men	RF-empirical	0.0320	-0.0045	0.0365	[-0.008, 0.152]	*
Men	GB-empirical	0.0320	-0.0052	0.0372	[-0.007, 0.153]	
Women	No correction	0.0024	0.0000	0.0024	[-0.007, 0.011]	
Women	Linear gap	0.0024	0.0011	0.0013	[-0.009, 0.011]	
Women	SC-entropy	0.0024	-0.0012	0.0036	[-0.008, 0.012]	
Women	RF-empirical	0.0024	0.0008	0.0016	[-0.007, 0.012]	*
Women	GB-empirical	0.0024	0.0027	-0.0003	[-0.010, 0.010]	
Crime 1	No correction	0.0214	0.0000	0.0214	[-0.006, 0.076]	
Crime 1	Linear gap	0.0214	0.0020	0.0195	[-0.007, 0.075]	
Crime 1	SC-entropy	0.0214	0.0186	0.0028	[-0.020, 0.051]	
Crime 1	RF-empirical	0.0214	-0.0048	0.0262	[-0.002, 0.081]	*
Crime 1	GB-empirical	0.0214	-0.0044	0.0258	[-0.002, 0.082]	
Crime 2	No correction	0.0178	0.0000	0.0178	[-0.008, 0.067]	
Crime 2	Linear gap	0.0178	0.0026	0.0152	[-0.009, 0.064]	
Crime 2	SC-entropy	0.0178	0.0163	0.0015	[-0.021, 0.045]	
Crime 2	RF-empirical	0.0178	-0.0031	0.0209	[-0.005, 0.070]	*
Crime 2	GB-empirical	0.0178	-0.0026	0.0205	[-0.004, 0.069]	
Crime 3	No correction	0.0120	0.0000	0.0120	[-0.010, 0.051]	
Crime 3	Linear gap	0.0120	0.0032	0.0088	[-0.013, 0.044]	
Crime 3	SC-entropy	0.0120	0.0122	-0.0002	[-0.022, 0.034]	
Crime 3	RF-empirical	0.0120	-0.0024	0.0144	[-0.006, 0.051]	*
Crime 3	GB-empirical	0.0120	-0.0028	0.0148	[-0.005, 0.052]	
Crime 4	No correction	0.0094	0.0000	0.0094	[-0.012, 0.047]	*
Crime 4	Linear gap	0.0094	0.0030	0.0063	[-0.015, 0.046]	
Crime 4	SC-entropy	0.0094	0.0094	-0.0000	[-0.021, 0.038]	
Crime 4	RF-empirical	0.0094	-0.0016	0.0110	[-0.010, 0.050]	
Crime 4	GB-empirical	0.0094	-0.0014	0.0108	[-0.010, 0.049]	

For fixed-method uncertainty we resample AIS, with replacement separately among five treated and seventeen control clusters, and refit the final 84/13-day calculation on each draw. Percentile intervals summarize 499 draws for each of 35 outcome–method estimates. The selected total-homicide Random Forest interval is $[-0.0126, 0.1552]$, and none of the seven selected fixed-method intervals excludes zero. For total homicides we additionally repeat the *entire* 154-window model-selection exercise inside each of 99 AIS draws, then refit that draw’s selected method in the strike window. The re-selection result is reported separately from fixed-method intervals. Only five treated AIS are available, so neither percentile construction nor model re-selection removes substantial few-cluster uncertainty; the 99-draw selection-aware

interval in particular has coarse tail resolution. AIS resampling also assumes that these clusters are a meaningful basis for uncertainty, not that treatment is randomized across them.

Table 6: Fixed-method versus full selection-aware AIS bootstrap for total homicides. Both rows share the original-sample point estimate; percentile intervals differ in whether selection is repeated.

Procedure	Draws	Estimate	95% interval	Re-selection
Random Forest fixed	499	0.0361	[-0.0126, 0.1552]	No
Full RMSE selection	99	0.0361	[-0.0254, 0.1433]	Yes

Across 99 selection-aware resamples, the pre-strike RMSE rule chose no correction 61, linear gap 0, Synthetic Control 19, Random Forest 15, Gradient Boosting 4. The observed-sample winner was Random Forest. Selection changes across draws, showing that the small placebo-RMSE advantage is sensitive to AIS composition. Neither interval is a randomization-based causal interval.

The selection frequencies in Table 6 describe this resampling exercise, not probabilities that a method identifies the true effect. No correction is selected most often despite the forest’s original-sample first place. The instability is consistent with a narrow performance gap and limited independent cluster information [Cawley and Talbot, 2010, Cameron and Miller, 2015, Ferman and Pinto, 2019].

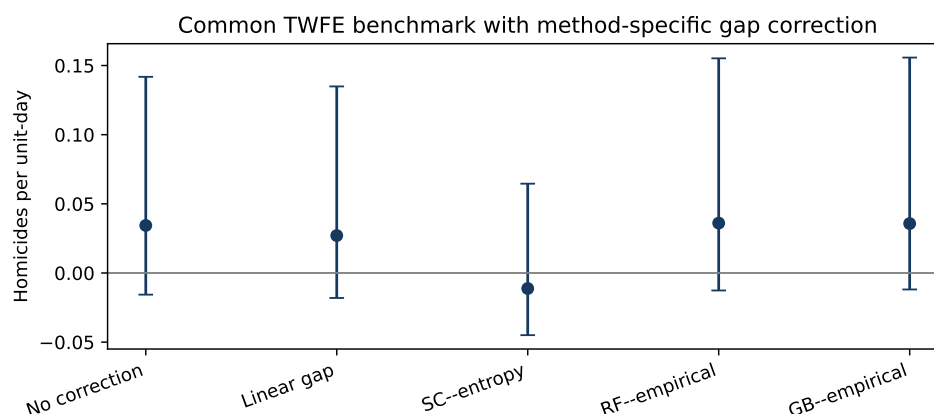


Figure 4: Total-homicide estimates and fixed-method 95% AIS-bootstrap percentile intervals, all aligned to the same TWFE benchmark. The displayed intervals do not repeat model selection; Table 6 gives the selection-aware comparison.

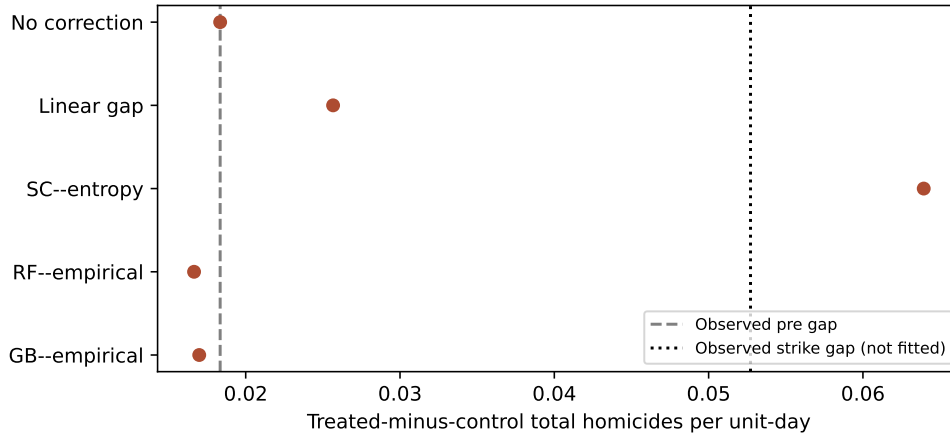


Figure 5: Total-homicide untreated strike-gap forecasts. The observed strike gap is displayed only for comparison and is excluded from every candidate’s fit and from the pre-strike selection.

6.5 External comparison and sensitivity

The released Mancha specification regresses the nonzero-observation unit–day outcome on a statewide strike indicator and its interaction with high gang exposure, absorbing municipality/district, AIS–year, month and weekday effects. It uses the 2014–2020 source sample rather than the balanced 97-day main window; in our replication it has 21,886 source unit–day observations after the original singleton/duplicate handling. Its interaction is not the unit/date TWFE DID in Equation 21: zero-filled days, date effects, sample duration, implicit observation weights and, for the published code, treatment coding all differ. With the executed four-AIS coding, the reproduced total-homicide interaction is 0.334996, matching the published 0.335 after rounding. Six other reproduced interactions are within 0.0006 of the printed coefficients; the largest discrepancy is for women (0.068401 versus published 0.069), which is not identical at three-decimal rounding and remains unresolved. Correcting that *same* external specification to five AIS gives 0.224398. Neither coefficient is inserted into Equation 22; the difference from this thesis’s 0.03437 TWFE is not an estimate of “bias corrected away.”

Table 8 changes the pre-period from 8 to 104 weeks, applies the executed four-AIS and mechanical six-AIS definitions, and aggregates at AIS–day rather than municipality/district–day. Four-AIS coding increases the 12-week TWFE to about 0.0568; adding AIS 07 produces about 0.0349. The AIS aggregate estimates are on a different count scale and weight AIS equally instead of municipalities. Synthetic Control’s negative main-window result is sensitive to hori-

Table 7: External replication of Mancha’s interaction regression versus this thesis’s distinct TWFE benchmark. Coefficients are not subtracted from one another.

Outcome	Published	Replicated four AIS	Corrected five AIS	Thesis TWFE
Total	0.335	0.335	0.224	0.034
Men	0.267	0.267	0.184	0.032
Women	0.069	0.068	0.040	0.002
Crime 1	0.297	0.297	0.236	0.021
Crime 2	0.267	0.267	0.188	0.018
Crime 3	0.180	0.180	0.083	0.012
Crime 4	0.159	0.159	0.060	0.009

zon, becoming slightly positive at 104 weeks. These contrasts show that treatment definition, forecast support and analysis unit matter; they are not alternative measurements of a known ATT.

Table 8: Total-homicide sensitivity to pre-period length, treatment definition, and aggregation. Entries are adjusted estimates, not comparable across the last unit-aggregation row in absolute scale.

Specification	TWFE	Linear	SC	RF	GB
Five AIS, 8 weeks	0.032	0.027	-0.011	0.037	0.036
Five AIS, 12 weeks	0.034	0.027	-0.011	0.036	0.036
Five AIS, 26 weeks	0.037	0.031	-0.011	0.036	0.036
Five AIS, 52 weeks	0.038	0.036	-0.011	0.035	0.036
Five AIS, 104 weeks	0.037	0.038	0.003	0.035	0.034
Four AIS, 12 weeks	0.057	0.050	0.008	0.061	0.061
Six AIS, 12 weeks	0.035	0.029	-0.019	0.038	0.037
AIS-day, 12 weeks	0.767	0.619	-0.754	0.790	0.785

Table 9: All-outcome pre-strike placebo performance (same 154 windows and models in each row).

Outcome	Method	Signed error	Variance	MAE	RMSE	Windows	Failures
Total	No correction	0.00028	0.000069	0.00675	0.00830	154	0
Total	Linear gap	-0.00008	0.000090	0.00755	0.00946	154	0
Total	SC-entropy	0.00105	0.000076	0.00718	0.00876	154	0
Total	RF-empirical	0.00202	0.000061	0.00642	0.00809	154	0
Total	GB-empirical	0.00018	0.000067	0.00655	0.00820	154	0
Men	No correction	0.00027	0.000060	0.00630	0.00775	154	0
Men	Linear gap	-0.00005	0.000082	0.00703	0.00904	154	0
Men	SC-entropy	0.00102	0.000069	0.00682	0.00840	154	0
Men	RF-empirical	0.00183	0.000055	0.00597	0.00767	154	0
Men	GB-empirical	0.00020	0.000063	0.00620	0.00793	154	0
Women	No correction	0.00001	0.000004	0.00140	0.00188	154	0
Women	Linear gap	-0.00003	0.000005	0.00171	0.00216	154	0
Women	SC-entropy	0.00034	0.000004	0.00148	0.00191	154	0
Women	RF-empirical	0.00033	0.000003	0.00138	0.00183	154	0

Outcome	Method	Signed error	Variance	MAE	RMSE	Windows	Failures
Women	GB-empirical	0.00004	0.000004	0.00154	0.00192	154	0
Crime 1	No correction	0.00006	0.000030	0.00446	0.00544	154	0
Crime 1	Linear gap	-0.00016	0.000037	0.00479	0.00610	154	0
Crime 1	SC-entropy	0.00057	0.000033	0.00463	0.00579	154	0
Crime 1	RF-empirical	0.00102	0.000028	0.00433	0.00538	154	0
Crime 1	GB-empirical	0.00003	0.000032	0.00450	0.00563	154	0
Crime 2	No correction	0.00006	0.000025	0.00411	0.00497	154	0
Crime 2	Linear gap	-0.00015	0.000031	0.00439	0.00559	154	0
Crime 2	SC-entropy	0.00056	0.000027	0.00423	0.00522	154	0
Crime 2	RF-empirical	0.00098	0.000022	0.00394	0.00483	154	0
Crime 2	GB-empirical	0.00007	0.000027	0.00413	0.00517	154	0
Crime 3	No correction	0.00006	0.000019	0.00352	0.00439	154	0
Crime 3	Linear gap	-0.00014	0.000025	0.00406	0.00504	154	0
Crime 3	SC-entropy	0.00056	0.000020	0.00357	0.00449	154	0
Crime 3	RF-empirical	0.00086	0.000018	0.00337	0.00428	154	0
Crime 3	GB-empirical	0.00003	0.000021	0.00363	0.00455	154	0
Crime 4	No correction	0.00008	0.000016	0.00318	0.00402	154	0
Crime 4	Linear gap	-0.00005	0.000022	0.00378	0.00474	154	0
Crime 4	SC-entropy	0.00056	0.000016	0.00324	0.00408	154	0
Crime 4	RF-empirical	0.00084	0.000016	0.00325	0.00413	154	0
Crime 4	GB-empirical	0.00011	0.000020	0.00350	0.00445	154	0

7 Discussion

The simulation supports a conditional case for correction, not a general preference for adjustment. Random Forest improves RMSE in S2–S5 but not in the parallel, weak-linear or high-noise cases. Synthetic Control and Gradient Boosting help in some stronger departures while leaving substantial residual bias. The implemented pooled Linear adjustment does not change TWFE at all; it should not be confused with the group-gap line fitted in the empirical study. Scenario-matched correction illustrates the value of correct specification, but its variance cost can outweigh its smaller bias. Its known-form advantage is unavailable to an investigator who does not know the untreated path. These results evaluate particular implementations under this design, rather than algorithm families in general.

The Ceará comparison demonstrates why predictive ranking and causal identification must remain separate. The forest’s narrow original-sample advantage is sensitive to AIS composition: full re-selection most often prefers no correction. This demonstrates selection instability, not which strike counterfactual is correct. Opposite-direction adjustments can share a TWFE

benchmark while embodying different forecasts. Moreover, the empirical forest uses redefined features, and Synthetic Control accesses contemporaneous donor outcomes. The ranking is therefore specific to complete procedures and information sets. Single-mean matching does not guarantee a credible untreated path.

Historical placebo performance does not establish that the untreated gap would obey the same relationship during a police strike. Overlapping training windows are not independent experiments. AIS resampling exposes sampling and selection variation within the observed cluster structure, but five treated clusters limit inference. Sex classification, the ambiguous AIS 07 cutoff and published-versus-executed treatment coding remain additional constraints.

The practical lesson is to state the estimating object before attempting a correction, align every forecast to its pre-period baseline and weights, pre-specify a no-correction candidate, select only with untreated pre-period evidence, and report the whole candidate set together with external estimands clearly separated. Such a workflow permits an adjustment without pretending that predictive validation identifies a missing causal counterfactual. Sensitivity analysis for nonparallel trends [Rambachan and Roth, 2023], richer donor matching or regularized augmentation [Abadie, 2021, Ben-Michael et al., 2021], and inference designed for very few treated clusters [Conley and Taber, 2011, MacKinnon and Webb, 2017] are relevant extensions, not procedures whose benefits have been demonstrated by the present calculations.

8 Conclusion

This thesis evaluates a transparent DID bias correction: predict the untreated treated–control gap during treatment, subtract its change from the same pre-treatment gap embedded in TWFE, and keep that benchmark common to all methods. In the controlled simulation, Random Forest provides the clearest broadly available improvements across the moderate and stronger violations, but loses that advantage in high noise. Correctly specified gap forecasts reduce average bias further without always minimizing RMSE. In the Ceará application, the original-sample pre-strike RMSE rule selects Random Forest for total homicides and five other outcomes, while retaining unadjusted TWFE for one. The selected total-homicide point estimate is close to TWFE, its

pre-strike RMSE improvement is modest, and complete AIS resampling often selects a different method. The selected-procedure total-homicide interval includes zero. These calculations compare historically predictive procedures for the differential exposure contrast, not directly observed strike-period ATT bias or the statewide strike effect. The value of the framework is its auditable link between estimand, forecast, validation, uncertainty and sensitivity, rather than a claim that one adjustment always dominates.

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